

# SAMUEL C MOGEN

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## EDUCATION

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### University of Colorado Boulder

Ph.D. in Atmospheric & Oceanic Sciences

Dissertation title: *Dynamics and Predictability of Marine Ecosystem Stressors*

Dissertation advisor: Nicole Lovenduski

M.S. in Atmospheric & Oceanic Sciences

Boulder, CO

May 2025

May 2022

### University of Virginia

B.S. in Environmental Science

Thesis advisor: Scott Doney

Charlottesville, VA

2020

## WORK EXPERIENCE

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### University of Colorado Boulder

Post-Doctoral Associate, Institute of Arctic and Alpine Research

Boulder, CO

2025-present

### University of Colorado Boulder

Graduate Research Assistant, Institute of Arctic and Alpine Research and Department of Atmospheric and Oceanic Sciences

Boulder, CO

2021-2025

### University of Virginia

Undergraduate Research Assistant, Computational Biogeochemistry Lab

Charlottesville, VA

2019-2020

### NOAA Pacific Marine Environmental Lab

NOAA Hollings Scholar, Physical Sciences Division

Seattle, WA

Summer 2019

## PUBLICATIONS

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In preparation:

9. **Mogen, S.C.**, Lovenduski, Capotondi, A., Bograd, S.J., Hazen, E.L., Pozo Buil, M., Jacox. *in preparation*. Ecosystem impacts of the poleward shift of the North Pacific Current . *JGR Oceans*.
8. **Mogen, S.C.**, Lovenduski, N.S., Clow, G.L. *in preparation*. Dynamics and predictability of marine phytoplankton extremes and compound events. *Geophysical Research Letters*.
7. Clow, G.L., Lovenduski, N.S., Levy, M.N., Lindsay, K., **Mogen, S.C.**, Kay, J.E. *in preparation*. Detectability of phytoplankton biomass extremes using simulated satellite chlorophyll observations. *Geophysical Research Letters*

In press:

6. Kohlmann, C., **Mogen, S.C.**, Cohen, J.T., Araujo, J., Beaudin, E., Bonino, G., Gagliardi, A., Lovenduski, N.S. *in press*. Surface and subsurface biogeochemical impacts of the 2019 Northeast Pacific marine heatwave. *Geophysical Research Letters*.

In print:

5. **Mogen, S.C.**, Lovenduski, N.S., Yeager, S.G., Capotondi, A., Bograd, S.J., Hazen, E.L., Pozo Buil, M., Jacox, M.G., Di Lorenzo, E., Kim, W., Rosenbloom, N.R.. (2024) Multi-month forecasts of marine heatwaves and ocean acidification extremes. *Nature Geoscience*, 17, 1261–1267.  
<https://doi.org/10.1038/s41561-024-01593-0>
4. **Mogen, S.C.**, Lovenduski, N.S., Yeager, S., Keppler, L., Sharp, J., Bograd, S.J., Quiros, N.C., Di Lorenzo, E., Hazen, E.L., Jacox, M.G., & Pozo Buil, M. (2023). Skillful Multi-Month Predictions of Ecosystem Stressors in the Surface and Subsurface Ocean. **Earth's Future**, 11(11).  
<https://doi.org/10.1029/2023ef003605>

3. Dukes, E., Cheng, S., **Mogen, S.C.**, Galloway, J., Leach, A., Trimble, A. R., Pettit, A., Compton, J., & Pennino, M. (2023). Footprints in Action: How UVA Is Managing Its Sustainability Stewardship. *Sustainability and Climate Change*, 16(1), 48–63.  
<https://doi.org/10.1089/scc.2022.0067>
2. **Mogen, S.C.**, Lovenduski, N.S., Dallmann, A.R., Gregor, L., Sutton, A.J., Bograd, S.J., Quiros, N. C., Di Lorenzo, E., Hazen, E.L., Jacox, M.G., Pozo Buil, M., & Yeager, S. (2022). Ocean Biogeochemical Signatures of the North Pacific Blob. *Geophysical Research Letters*, 49(9).  
<https://doi.org/10.1029/2021gl096938>
1. Yeager, S.G., Rosenbloom, N., Glanville, A.A., Wu, X., Simpson, I., Li, H., Molina, M.J., Krumhardt, K., **Mogen, S.C.**, Lindsay, K., Lombardozzi, D., Wieder, W., Kim, W.M., Richter, J.H., Long, M., Danabasoglu, G., Bailey, D., Holland, M., Lovenduski, N.S., Strand, W., & King, T. (2022). The Seasonal-to-Multiyear Large Ensemble (SMYLE) prediction system using the Community Earth System Model version 2. *Geoscientific Model Development*, 15(16), 6451–6493.  
<https://doi.org/10.5194/gmd-15-6451-2022>

## HONORS AND AWARDS

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|                                                                        |           |
|------------------------------------------------------------------------|-----------|
| Community Earth System Model Graduate Student Award                    | 2024      |
| Atmospheric and Oceanic Sciences Departmental Service Award            | 2020-2025 |
| NSF Graduate Research Fellowship, Honorable Mention                    | 2021      |
| Atmospheric and Oceanic Sciences Departmental Teaching Assistant Award | Fall 2020 |
| Ernest F. Hollings Scholar, NOAA                                       | 2018      |

## SERVICE & OUTREACH

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|                                                                                                                                                                                             |           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| Peer-Reviewer for: <i>Geophysical Research Letters</i> , <i>Earth's Future</i> , <i>Climate Dynamics</i> , <i>Journal of Geophysical Research: Oceans</i> , <i>Progress in Oceanography</i> |           |
| Co-Lead and Co-Developer, Department Microaggression and Bias Seminars                                                                                                                      | 2021-2025 |
| Co-Lead and Organizer, ATOC REU Python Bootcamp                                                                                                                                             | 2021-2024 |
| Chair, Justice, Equity, Diversity and Inclusion Committee, ATOC                                                                                                                             | 2020-2025 |

## SKILLS & INTERESTS

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|                             |                                                       |
|-----------------------------|-------------------------------------------------------|
| <b>Computer Languages</b>   | python, MATLAB, R, shell scripting                    |
| <b>Data &amp; Databases</b> | NetCDF, ZARR                                          |
| <b>Design</b>               | L <sup>A</sup> T <sub>E</sub> X, Adobe Creative Cloud |

## SELECTED GRADUATE COURSEWORK

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|                                       |                                |
|---------------------------------------|--------------------------------|
| Biogeochemical Oceanography           | Atmospheric Thermodynamics     |
| Introduction to Physical Oceanography | Intro to Atmospheric Radiation |
| Dynamics of the Atmosphere and Oceans | Global Carbon Cycle            |
| Advanced Ocean Dynamics               | Objective Data Analysis        |
| Chaos and Predictability              |                                |

## SELECTED PRESENTATIONS

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Invited Talks:

4. **Mogen, S.** Extremes and predictability of ocean biogeochemical ecosystem stressors. NOAA Physical Sciences Laboratory Seminar. February, 2025.
3. **Mogen, S.** Multi-month forecasts of marine heatwaves and ocean acidification extremes. CESM Workshop. June, 2024.
2. **Mogen, S.** Skillful Multi-month Predictions of Ecosystem Stressors in the Surface and Subsurface Ocean. BGC Argo Science Meeting. November, 2023.

1. **Mogen, S.** Near-term predictions of (sub)surface biogeochemical state. BGC Argo Science Meeting. September, 2022.

Conference Talks:

14. **Mogen, S,** Lovenduski, NS, Yeager, S, Capotondi, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Kim, W, Rosenbloom, NR. "Multi-month forecasts of marine heatwaves and ocean acidification extremes." OceanPredict Symposium 2024. Paris, France. November, 2024.
13. **Mogen, S,** Lovenduski, NS, Yeager, S, Capotondi, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Kim, W, Rosenbloom, NR. "Multi-month forecasts of marine heatwaves and ocean acidification extremes." Ocean Sciences Meeting. New Orleans, LA. February, 2024.
12. **Mogen, S,** Lovenduski, NS, Yeager, S. "Multi-month predictions of extremes in ocean temperature and acidity." ICTP/CLIVAR Summer School on Marine Heatwaves. Trieste, Italy. July, 2023.
11. **Mogen, S,** Lovenduski, NS, Yeager, S, Keppler, L., Sharp, J., Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Predicting marine stressors multiple months in advance." SOLAS Summer School. Mindelo, Cape Verde. June, 2023.
10. **Mogen, S,** Lovenduski, NS, Yeager, S, Keppler, L, Sharp, J, Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Predicting marine stressors multiple months in advance . " NOAA/CPO/CVP Webinar Series - Decadal Variability and Predictability. Virtual. June, 2023.
9. **Mogen, S,** Lovenduski, NS, Yeager, S, Keppler, L, Sharp, J, Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Skillful predictions of multiple marine stressors in the surface and subsurface ocean." CESM Ocean Model Working Group Meeting. Boulder, CO. February, 2023.
8. **Mogen, S,** Lovenduski, NS, Yeager, S, Keppler, L, Sharp, J, Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Near-term predictions of (sub)surface marine stressors." 5th International Symposium on the Ocean in a High CO2 World. Lima, Peru. September, 2022.
7. **Mogen, S,** Lovenduski NS, Yeager, S. "Multi-Month Predictions of Ocean Acidification in CESM SMYLE." CESM Earth System Prediction Working Group Meeting. Boulder, CO. February, 2022.
6. **Mogen, S,** Lovenduski NS, Yeager, S, Dallman, A, Gregor, L, Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Ocean biogeochemical signatures of the North Pacific Blob." ATOC Departmental Colloquium. Boulder, CO. February, 2022.
5. **Mogen, S,** Lovenduski NS, Yeager, S, Dallman, A, Gregor, L, Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Ocean biogeochemical signatures of the North Pacific Blob." Ocean Sciences Meeting. Virtual. February, 2022.
4. **Mogen, S,** Lovenduski NS, Yeager, S, Dallman, A, Gregor, L, Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Ocean biogeochemical signatures of the North Pacific Blob." WCRP Workshop on Extremes in Climate Prediction Ensembles (ExCPens). Virtual. October, 2021.
3. **Mogen, S,** Lovenduski NS, Yeager, S, Dallman, A, Gregor, L, Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Ocean biogeochemical signatures of the North Pacific Blob." Graduate Climate Conference. Virtual. October, 2021.
2. **Mogen, S,** Lovenduski NS, Yeager, S, Dallman, A, Gregor, L, Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. "Ocean biogeochemical signatures of the North Pacific Blob." Eastern Pacific Ocean Conference. Fallen Leaf Lake, CA. September, 2021.

1. **Mogen, S**, Pilcher, D, Cross, JN. “Outlook on Changes in Bering Sea Oxygen Cycling.” Hollings Scholar Symposium. Silver Spring, MD. August, 2019.

Conference Posters:

7. **Mogen, S**, Lovenduski NS, “Multi-month forecasts of marine heatwaves and ocean acidification extremes.” AGU. Washington, D.C. December, 2024.
6. **Mogen, S**, Lovenduski NS, “Multi-month forecasts of marine heatwaves and ocean acidification extremes.” Earth System & Space Science (ESSS) Poster Session: Boulder, CO. December, 2023.
5. **Mogen, S**, Lovenduski, NS, Yeager, S, Keppler, L., Sharp, J., Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. “Skillful predictions of multiple surface and subsurface marine stressors.” SOLAS Summer School. Mindelo, Cape Verde. June, 2023.
4. **Mogen, S**, Lovenduski, NS, Yeager, S, Keppler, L., Sharp, J., Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. “Skillful predictions of multiple surface and subsurface marine stressors.” American Meteorological Society Meeting. Denver, CO. January, 2023.
3. **Mogen, S**, Lovenduski, NS. “Predictions of multiple ocean stressors in CESM-SMYLE in Coral Reef Regions.” US Clivar Workshop on Societally-Relevant Multi-Year Climate Predictions. Boulder, CO. March, 2022.
2. **Mogen, S**, Lovenduski NS, Dallman, A, Gregor, L, Sutton, A, Bograd, S, Hazen, E, Pozo Buil, M, Jacox, M, Di Lorenzo, E, Cordero Quiros, N. ”Ocean biogeochemical signatures of the North Pacific Blob.” Earth System Space Science (ESSS) Poster Session: Boulder, CO. December, 2021.
1. **Mogen, S**, Doney, S, Pilcher, D, Cross, JN. “Outlook on Changes in Bering Sea Oxygen Cycling.” Ocean Sciences Meeting: San Diego, CA. February 2020.